Open and FAIR Training Materials: Practices and Resources from SIB/ELIXIR-CH

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Why go open and FAIR?

Sharing educational is a valuable practice that benefits teachers, learners, and the broader life sciences community. Openly shared materials enable:

- **Reuse**, reducing duplication of effort in content creation
- **Community building**, fostering collaboration and quality improvement through collective feedback
- **Broader accessibility**, allowing learners worldwide to acquire skills on demand
- **Improved organization**, as preparing materials for sharing encourages clarity and modular design.

Originally developed for data, the FAIR principles can also be applied to training materials.

FAIRify Training Material at SIB

We address this challenge by:

- Setting up a dedicated **FAIR Training Task Force at SIB** to develop and promote best practices.
- Developing **glittr.org**, a curated database indexing GitHub-hosted training materials with standardized metadata, to help you find and compare bioinformatics training material.
- Advocating for **open licenses (e.g., CC-BY)** to maximize potential reuse.
- Producing a **practical flow chart** to guide creators in FAIRifying their training material (Figure 1). This workflow emphasizes the importance of licensing, standardizing metadata, and community collaboration to maximize the impact of educational materials.
- Supporting trainers **from planning and developing content** in open formats, such as Markdown or Jupyter notebooks, **to publishing their training materials** on platforms such as GitHub or Zenodo (which provides persistent identifiers in the form of DOIs).

In this way, our trainers contribute to a sustainable ecosystem of training resources that support global, equitable access to high-quality bioinformatics education.

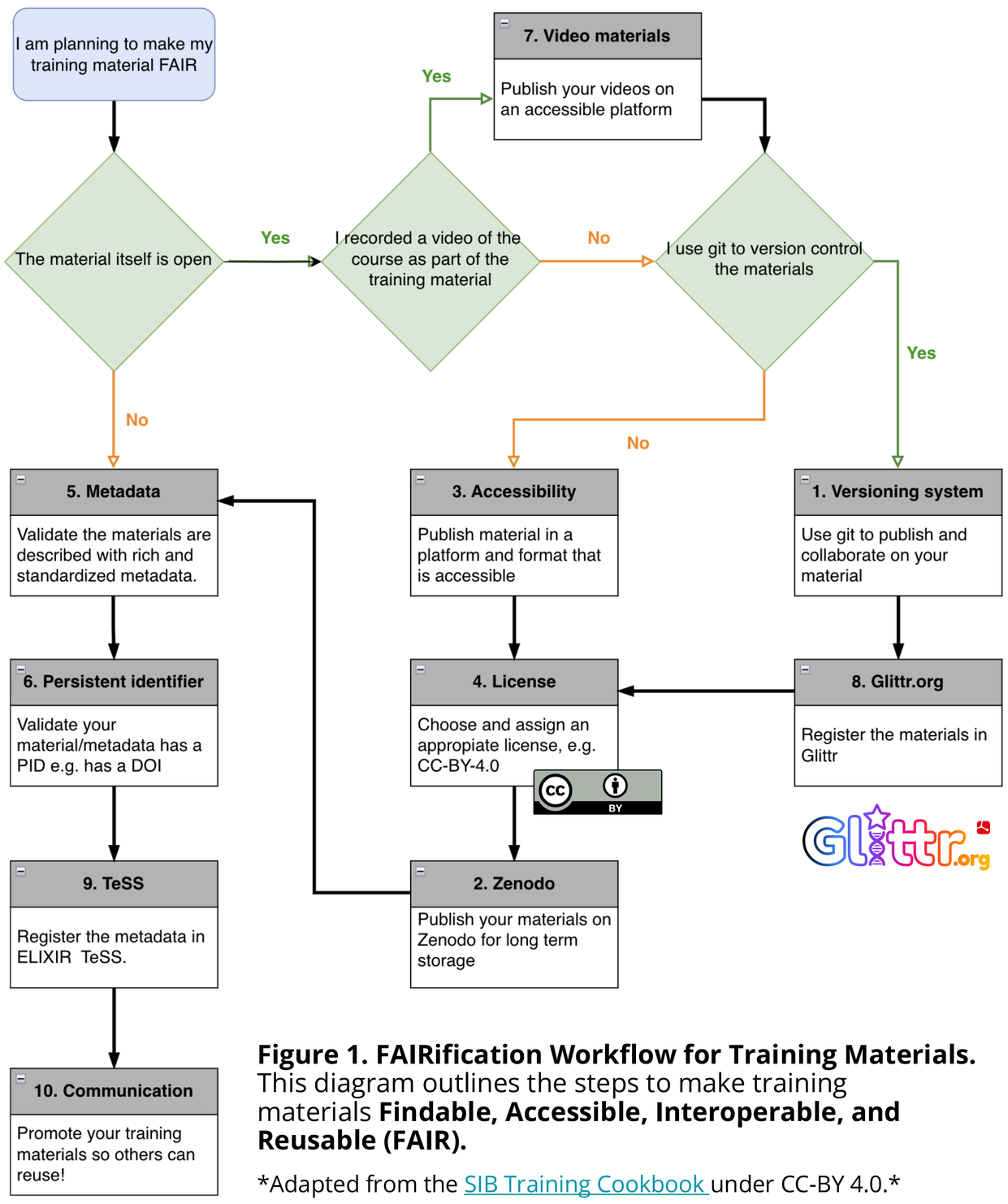

Figure 1. FAIRification Workflow for Training Materials. This diagram outlines the steps to make training materials **Findable, Accessible, Interoperable, and Reusable (FAIR).**

Adapted from the SIB Training Cookbook under CC-BY 4.0.

Empowering a Global Training Community

Our goal is to **catalyse FAIR training practices** within SIB, national and international institutions.

By following the Training Platform guidelines[1,2] and integrating FAIR practices with community platforms, we create sustainable, adaptable resources for life sciences education.

Our initiatives align with **SIB's commitment to open science and capacity building**, ensuring materials are not only FAIR but also pedagogically sound and adaptable to diverse learning contexts.

[1] Garcia L, Batut B, Burke ML, Kuzak M, Psomopoulos F, Arcila R, et al. (2020) Ten simple rules for making training materials FAIR. PLoS Comput Biol 16(5): e1007854.
[2] Alexandra Holinski, Alexia Cardona, Alice Matimba, Allegra Via, Anastasios Anastasios Papaioannou, Anna Swan, Bruna Piereck, et al. ELIXIR Fair Training Handbook. Zenodo, March 19, 2024. https://doi.org/10.5281/ZENODO.8239503.

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